



SUKKUR IBA UNIVERSITY

DEPARTMENT OF COMPUTER SYSTEMS ENGINEERING

FINAL YEAR DESIGN PROJECT PROPOSAL SYNOPSIS

Group Members	Supervisor	Project Title	Area of Field
1. M. Usman 2. Asma Channa 3. Abdul Moiz	Dr. Abdul Sattar Chan	SafeDrive AI: Intelligent Dashboard Device for Driver Anomaly Detection	• Computer Vision • Machine Learning • Embedded Systems
	Co-Supervisor		
	Dr. Junaid Bhatti		
Project Synopsis (200 Words Max)			
Road safety is a growing concern due to the increasing number of accidents caused by driver fatigue, distraction, and inattention. SafeDrive AI aims to address this issue through a smart, embedded monitoring system designed to detect abnormal driver behavior in real time. Installed on the vehicle's dashboard, the system uses a combination of computer vision, machine learning, and sensor integration to track key indicators such as eye closure, head movement, facial orientation, and physical driving patterns. Upon identifying anomalies like drowsiness, inattentiveness, or erratic motion the system immediately generates an alert and logs the event for further review. The device is compact, energy-efficient, and designed with affordability and scalability in mind, making it suitable for both personal and commercial use. By intervening before a potential incident occurs, SafeDrive AI promotes responsible driving and contributes to safer roads through proactive driver assistance.			
Possible Deliverable of the Project			
The deliverable of this project is to design and develop SafeDrive AI, a smart embedded system for detecting and alerting anomalies in driver behavior to improve road safety. In order to achieve this, the following are the key sub-goals: 1. Implement real-time driver monitoring using computer vision techniques with pre-trained convolutional neural networks (CNNs) for facial landmark detection, eye-tracking, and drowsiness recognition. 2. Apply multi-sensor data fusion using anomaly detection algorithms and deep learning to enhance behavioral analysis accuracy. 3. Develop a real-time feedback system that triggers alerts based on predictions from deep learning classifiers trained to identify risky or inattentive driving behavior. 4. Create a lightweight interface or application that visualizes model outputs and logs anomaly data using Backend support for model inference and result interpretation.			
FYP Committee Comments			

Signature of Supervisor

Signature of Co-Supervisor

Signature of FYP-Coordinator